HUNG-CHI, TSAI

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Education

•Feng Chia University *Jul. 2022 - Aug. 2024*
Master of Science in Computer Science and Information Engineering Taichung, Taiwan

•Providence University *Sep. 2018 - Jun. 2022*
Bachelor of Science in Computer Science and Information Engineering Taichung, Taiwan

Experience

•AI New Talent Program, Ministry of Economic Affairs *Oct. 2025 – Present*
Cloud AI Architect Trainee (Hosted by NYCU) Hsinchu City, Taiwan

- Learned the full lifecycle of AI systems, from hardware inference acceleration to cloud-based deployment.
- Implemented LLM fine-tuning, inference acceleration techniques, and developed RAG (Retrieval-Augmented Generation) architectures.
- Utilized Azure cloud infrastructure with Docker and GitHub Actions to establish CI/CD pipelines for automated AI model deployment.
- Acquired practical experience in Linux system administration and network planning (CCNA standards) to ensure service stability and security.

Personal Projects

Location Privacy-Preserving Parking Recommendation Based on Federated Learning [*1] *Feb. 2023 - Aug. 2024*
Master's Thesis

* **System Development**: Developed a privacy-preserving parking recommendation system integrating Federated Learning to ensure decentralized data processing.
* **Privacy Mechanism**: Implemented spatio-temporal obfuscation to effectively protect sensitive location information while maintaining high recommendation accuracy.
* **Model Optimization**: Employed Long Short-Term Memory (LSTM) deep learning models for parking prediction based on long-term temporal dependencies in historical data.
* **Academic Achievement**: Research findings presented at the APWCS 2024 international conference. This project was supported by the National Science and Technology Council (NSTC) under grant 112-2221-E-194-048-MY2.

Smart Traffic Assistant for Visually Impaired [*2] *Sep. 2020 - Jan. 2022*
Undergraduate Project

* **Model Development**: Developed a high-precision YOLOv4 traffic recognition model using Python and OpenCV for image pre-processing and annotation.
* **Computer Vision:**:Implemented real-time Object Detection on NVIDIA Jetson TX2 NX to identify signal status and countdown timers.
* **System Integration**:Designed an automated audio feedback system to provide users with precise voice prompts for safe street crossing.
* **Impact**:Delivered a portable, camera-based solution that empowers visually impaired users to travel independently, overcoming noise-complaint issues of traditional signals.

Technical Skills

Python: Developed Python-based privacy-preserving systems using Federated Learning and LSTM, and implemented YOLOv4 models.
Git: Collaborated with team members using Git for version control, managing code integration through branching, merging, and resolving conflicts to ensure project stability.
Spring Boot: Developed full-stack systems with a modular three-tier architecture, implementing RESTful APIs and securing financial data against SQL Injection and XSS.
Web Development: Built responsive user interfaces using Vue.js and JavaScript, integrating frontend components with backend services via Axios.
C Programming: Self-studying basic syntax and logic control; currently focusing on memory management and pointer operations to enhance low-level programming skills.

Achievements

–**Microsoft Certified: Azure AI Fundamentals (AI-900),** Microsoft *2026*
–**Honorable Mention**[*1]**,** The 19th Workshop on Wireless, Ad Hoc and Sensor Networks (WASN) *2024*
–**Bronze Medal**[*2]**,** Undergraduate Project Competition, College of Computing and Informatics, PU *2022*
–**ICDL Advanced Certificates: Word 2016 & Excel 2016,** ICDL Foundation *2022*